

## Chakrayuddh Kimsovanna

Graduate Student | Economics & Social Data Analytics | R, Stata, SQL

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Graduate student in Social Data Analytics and Research with training in economics, econometrics, and data management. I am interested in development economics, political economy, and applied economic research, and my projects have involved data cleaning, panel analysis, model diagnostics, database work, and empirical writing.

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## EDUCATION

### The University of Texas at Dallas

*M.S. in Social Data Analytics and Research* | Expected Summer 2026 | GPA: 3.945

### The University of Texas at Dallas

*B.S. in Economics, Minor in International Political Economy* | December 2024 | GPA: 3.781 | Cum Laude

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## MAIN PROJECTS

### Dollarization and Foreign Direct Investment in Latin America and the Caribbean

May 2026

Paper: [\[Link\]](#) | GitHub: [\[Link\]](#)

- Built a country-year panel for Latin American and Caribbean economies using IADB dollarization data and World Bank indicators pulled through R.
- Cleaned and merged the data, handled missing observations, created one-year lags, and compared fixed effects and random effects specifications.
- Ran diagnostics for heteroscedasticity, serial correlation, and cross-sectional dependence; estimated final models with AR(1) errors and country-clustered robust standard errors.
- Found no significant relationship between dollarization and FDI inflows, while trade openness remained positive and significant.

### H-1B Visa Sponsorship Trends in Finance-Related Occupations, 2016-2024

March-May 2025

Paper: [\[Link\]](#) | GitHub: [\[Link\]](#)

- Cleaned and combined large USCIS and Department of Labor H-1B datasets to study finance-related job petitions from 2016 to 2024.
- Filtered messy job-title text with custom finance-related keywords, standardized case status fields, removed unusable entries, and converted files into RDS format for more efficient processing.
- Used text mining methods, including sentence embeddings and K-means clustering, to group finance-related job titles into eight categories, including banking/credit, risk management, tax/audit/accounting, analytics, portfolio management, and compliance.
- Found that tax, audit, and accounting roles made up the largest share of finance-related petitions, while certification rates varied more by policy period than by finance subfield.

### Cambodia Foreign Direct Investment and Trade Database Application

March-May 2025

Interactive App: [\[Link\]](#) | GitHub: [\[Link\]](#)

- Built a database application to organize Cambodia's FDI and trade data, using Open Development Cambodia for investment projects and World Bank WITS for trade data.
- Cleaned and checked project-level FDI records manually, especially when company names, investor countries, or project ownership were unclear.
- Designed the database around many-to-many relationships between FDI projects, companies, investor countries, and provinces, since many projects involved multiple firms or locations.
- Converted the PostgreSQL schema into SQLite/RSQLite so the database could run inside an R Shiny application and be published online.
- Added searchable tables, filters, import-export charts, and a Leaflet map to make the data easier to explore by country, sector, province, and product type.

### From Crisis to Control: Cambodia's Responses in Fighting Malaria

April 2024

Paper: [\[Link\]](#)

- Looked at why malaria persisted in Cambodia even as the country made progress, especially in poorer and remote rural areas where healthcare access remained limited.
- Wrote about how malaria affected poorer households through missed work, missed school, treatment costs, and the need to borrow money or sell land/livestock to afford care.

- Discussed the country's response to malaria, including ACT treatment, help from international NGOs, awareness campaigns, mosquito nets, and breeding guppy fish in standing water to reduce mosquito larvae.

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## PRESENTATIONS

- "H-1B Visa Sponsorship Trends in Finance-Related Occupations, 2016-2024," Graduate Research Project Presentation, University of Texas at Dallas, 2025.
- "Designing a Database for Cambodia's Foreign Direct Investment and Trade," Graduate Research Project Presentation, University of Texas at Dallas, 2025.
- "Bangladesh-Myanmar Relations Regarding the Rohingya Refugee Crisis," Undergraduate Research Presentation, University of Texas at Dallas, 2022.

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## OTHER WRITING

### Policy Recommendations to Improve the Quality of Life for Afghan Women

April 2024

Policy Memo: [\[Link\]](#)

- Co-wrote a policy memo on ways international organizations could support Afghan women under Taliban rule, especially through education access, private-sector work, local leadership, and international advocacy.

### Bangladesh-Myanmar Relations Regarding the Rohingya Refugee Crisis

December 2022

Paper: [\[Link\]](#)

- Wrote a research essay on the Rohingya refugee crisis, focusing on Myanmar's persecution of Rohingya Muslims, Bangladesh's burden in hosting refugees, and the social, economic, and diplomatic tensions created by failed repatriation efforts.

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## SKILLS & ADDITIONAL INFORMATION

- **Technical Skills:** R, Stata, SQL/PostgreSQL, SQLite, Excel, LaTeX/R Markdown, GitHub, Python
- **Research Methods:** Regression analysis, panel data methods, model diagnostics, robustness checks, data cleaning, data visualization
- **Research & Writing Tools:** Microsoft Word, PowerPoint, Google Docs/Sheets/Slides
- **Relevant Coursework:** Econometrics, Regression & Multivariate Analysis, Descriptive & Inferential Statistics, Mathematical Economics, Business & Economic Forecasting, Knowledge Mining, Information Management

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## CERTIFICATIONS

- Certificate in Data Science and Business Analytics – *The University of Texas at Austin, McCombs School of Business* (May 2024); included applied projects in data analysis, machine learning, clustering, model tuning, and predictive modeling in Python.

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## AWARDS & HONORS

- Cum Laude, B.S. in Economics, University of Texas at Dallas, 2024
- Dean's List, Fall 2024